PRODUCT BRIEF

BuddyMatcher Introduction and User Guide

BuddyMatcher helps program organizers create high-quality one-to-one cultural exchange pairs by combining survey insights, profile context, and practical administration tools in one workflow.

This guide introduces the main screens and explains how users and admins can operate the platform effectively.

Language: English

1. What BuddyMatcher Does

BuddyMatcher collects participant profile and survey data, computes compatibility, and presents each user with their recommended buddy.

Structured Onboarding

Participants provide public profile details and private matching answers.

Compatibility-Driven Matching

Pairing uses weighted survey dimensions and forced-choice preferences.

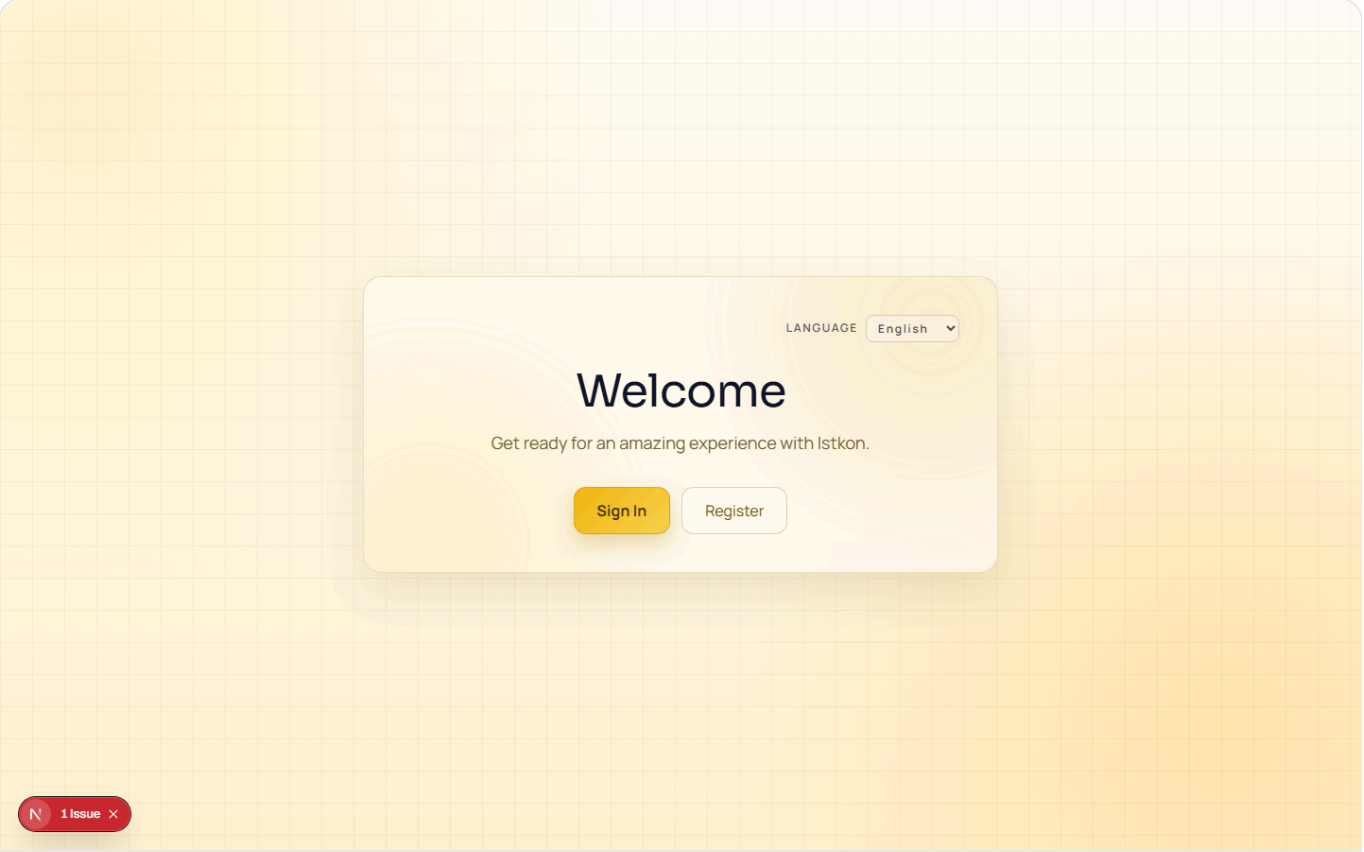
Participant Directory

All participants can browse public profiles and social links.

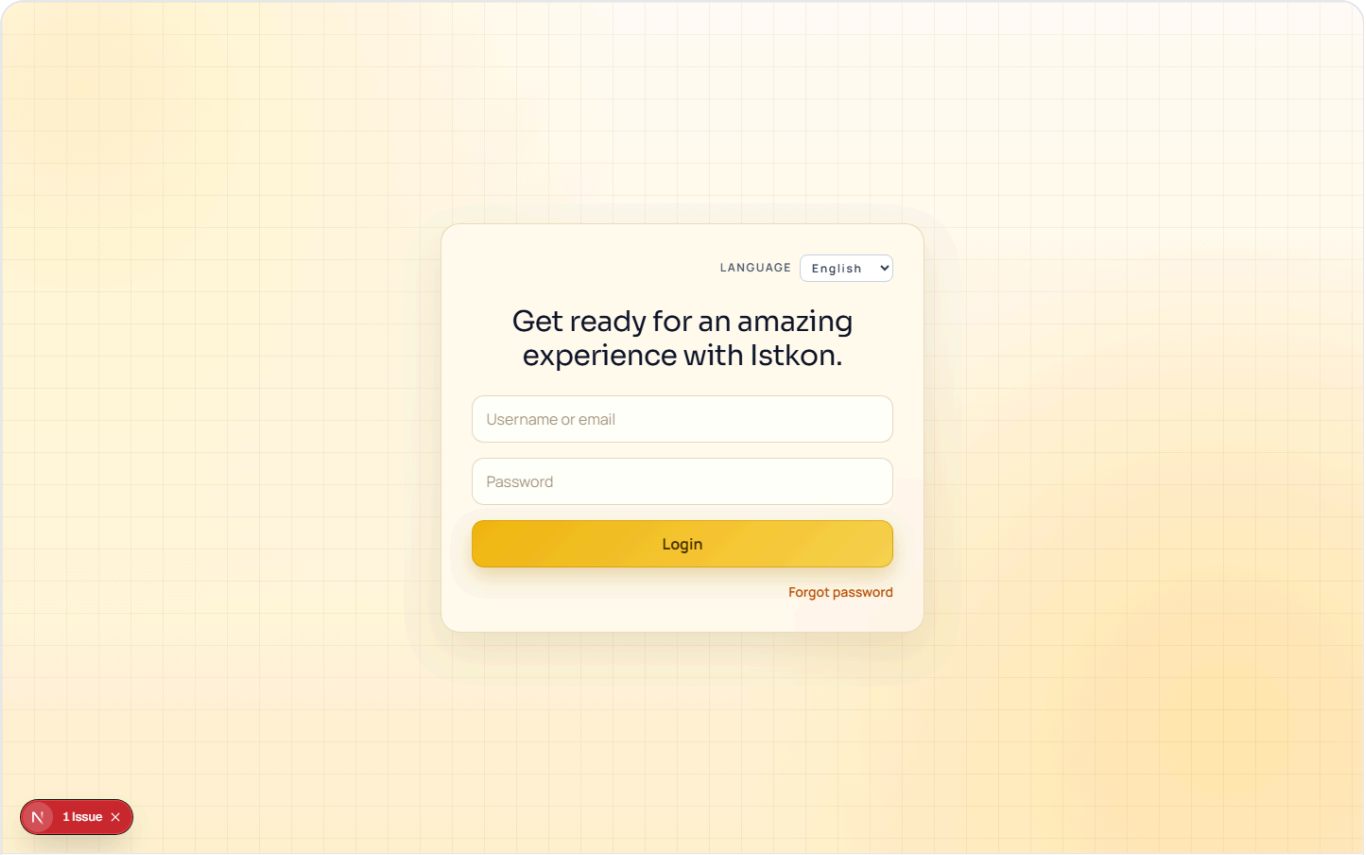
Admin Controls

CSV import, answer permissions, and match runs are centralized.

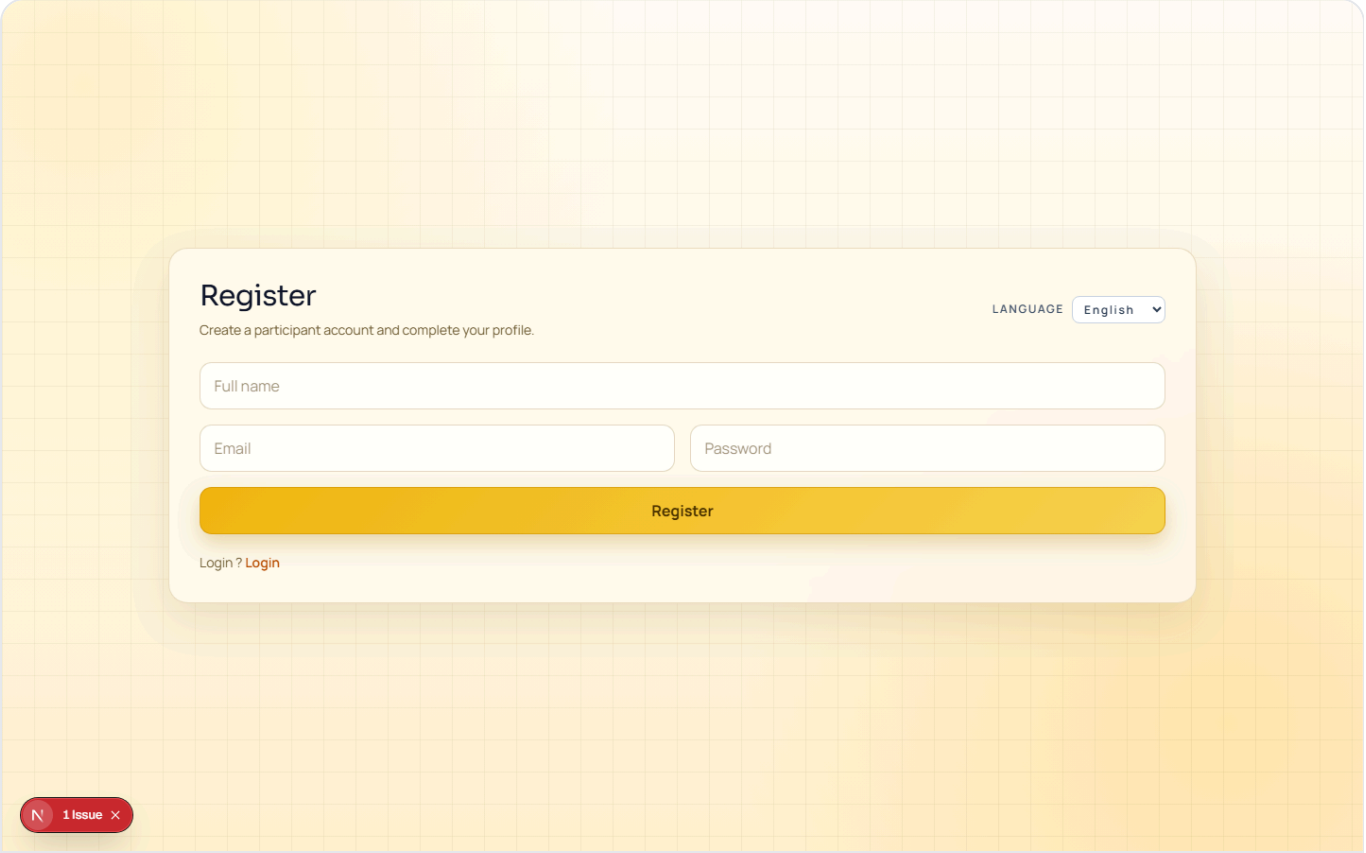
2. Main Screens



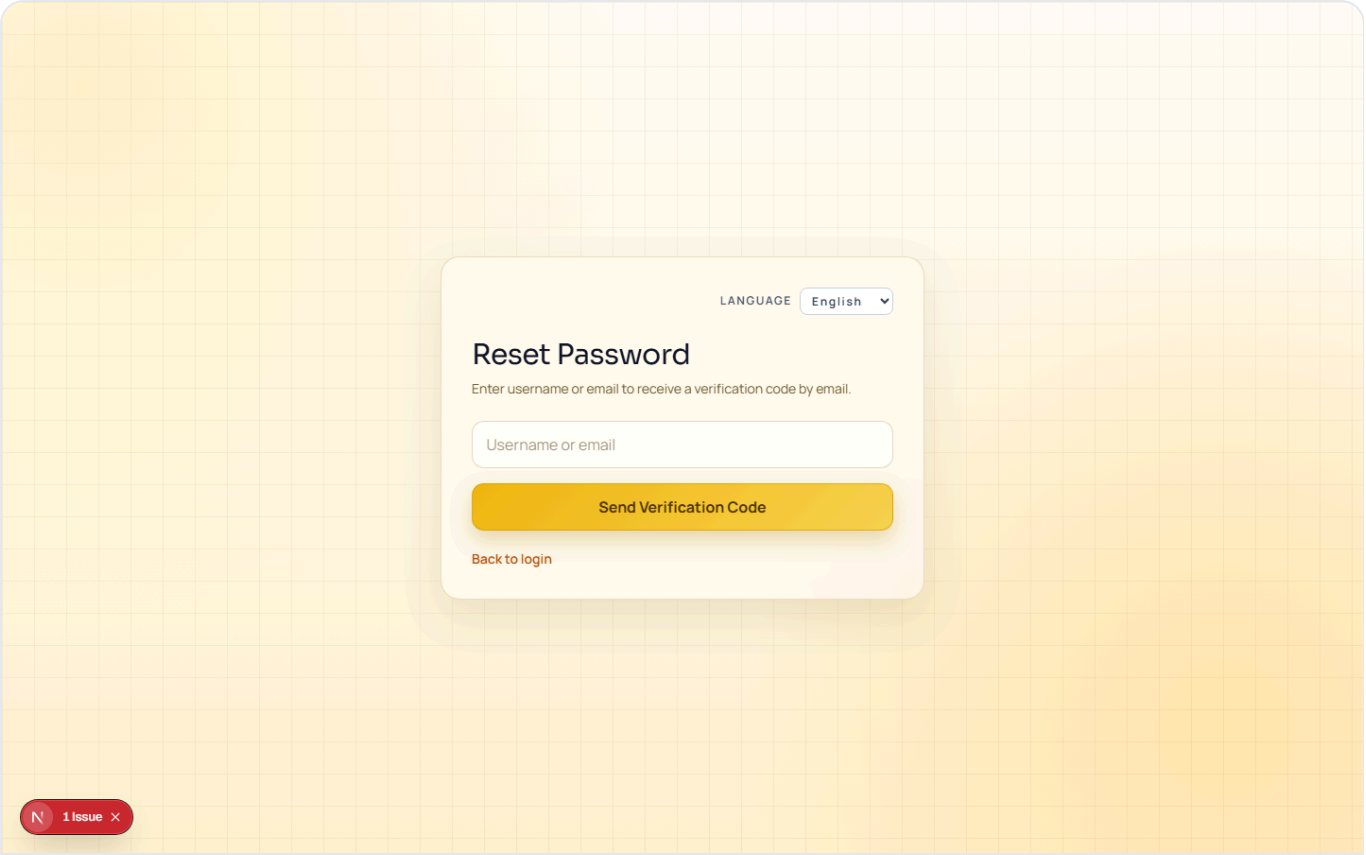
Home page with clear navigation to login and registration.



Login page where users sign in with username/email and password.

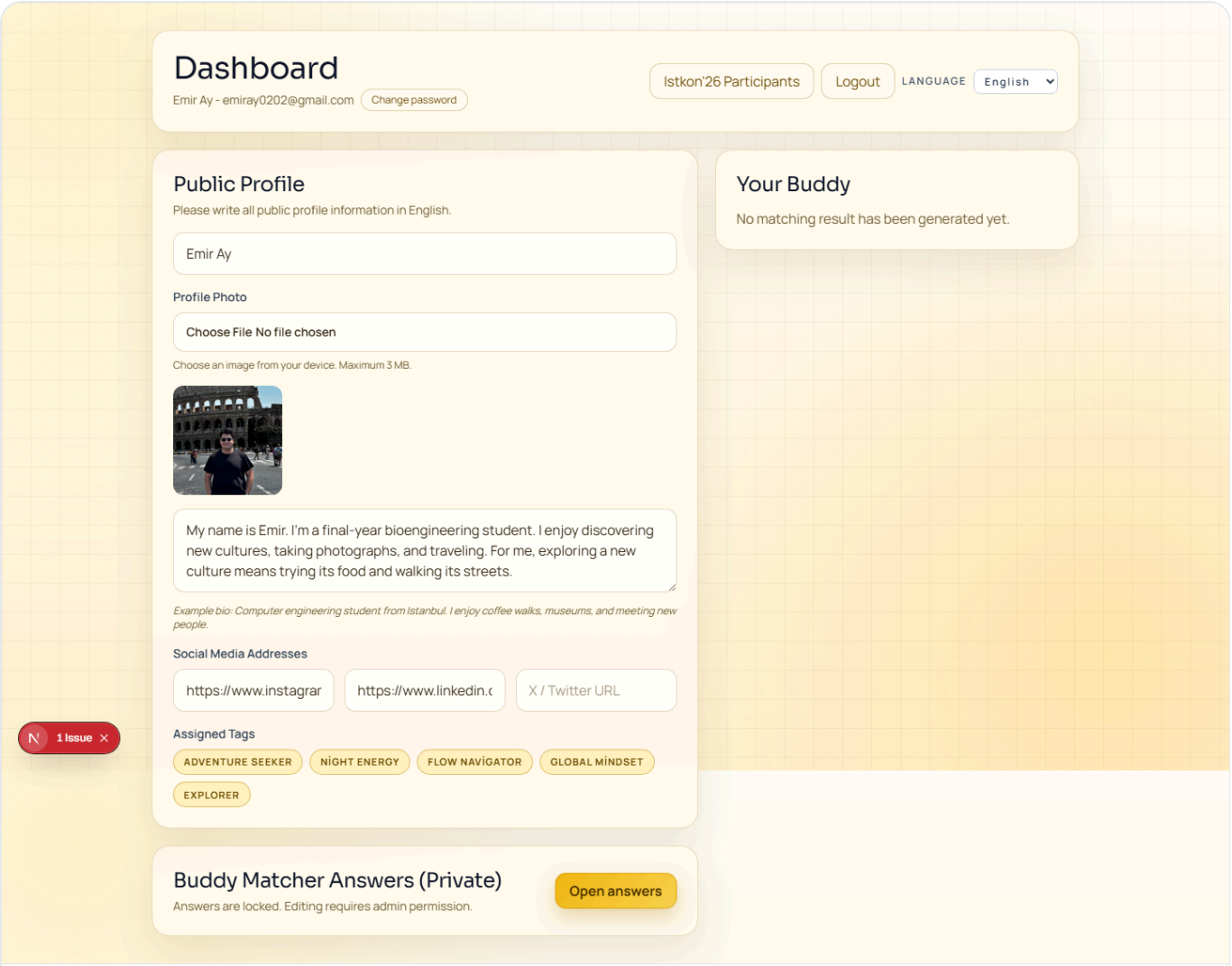


Registration screen for new participant account creation.

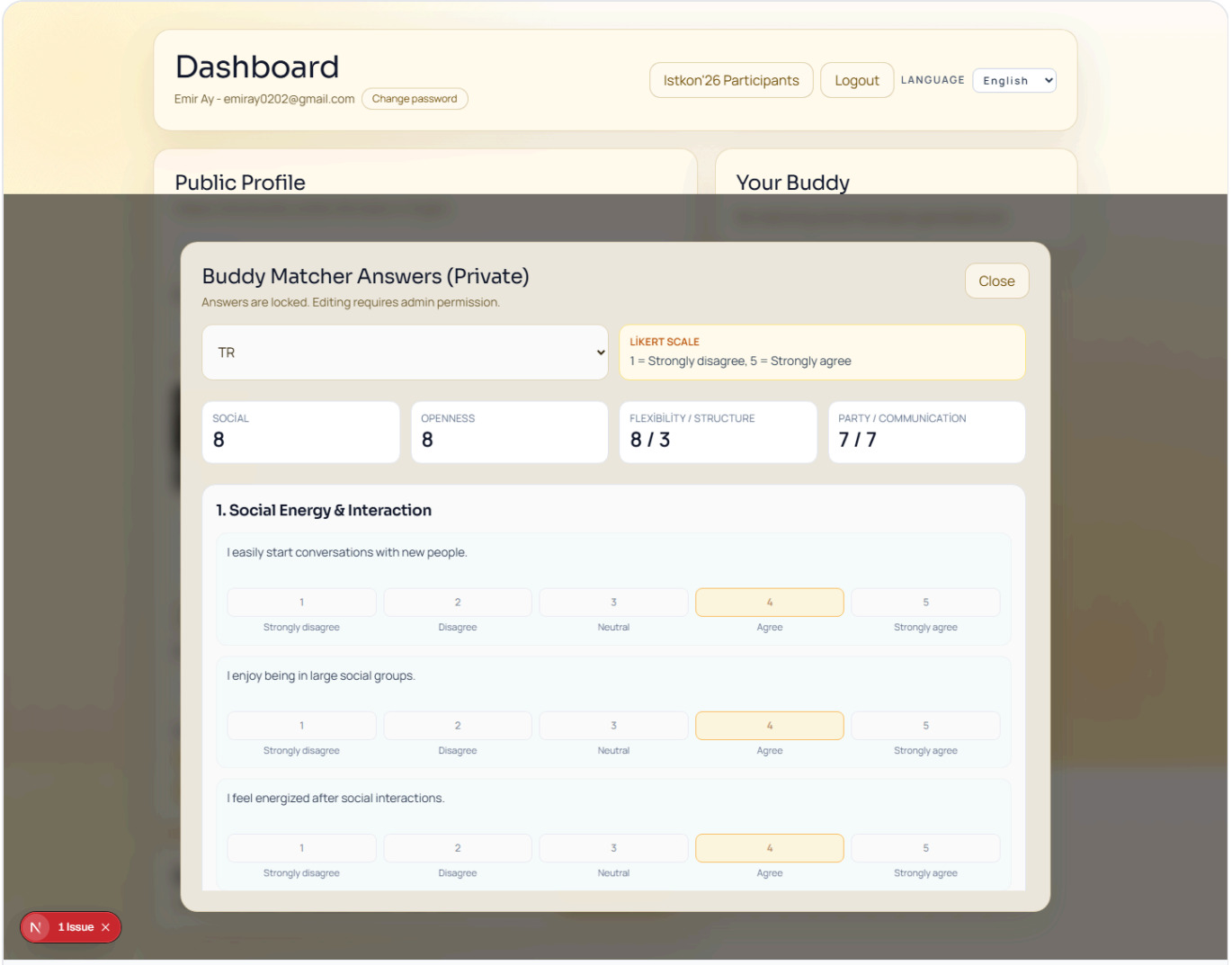


Password recovery flow with verification code delivery.

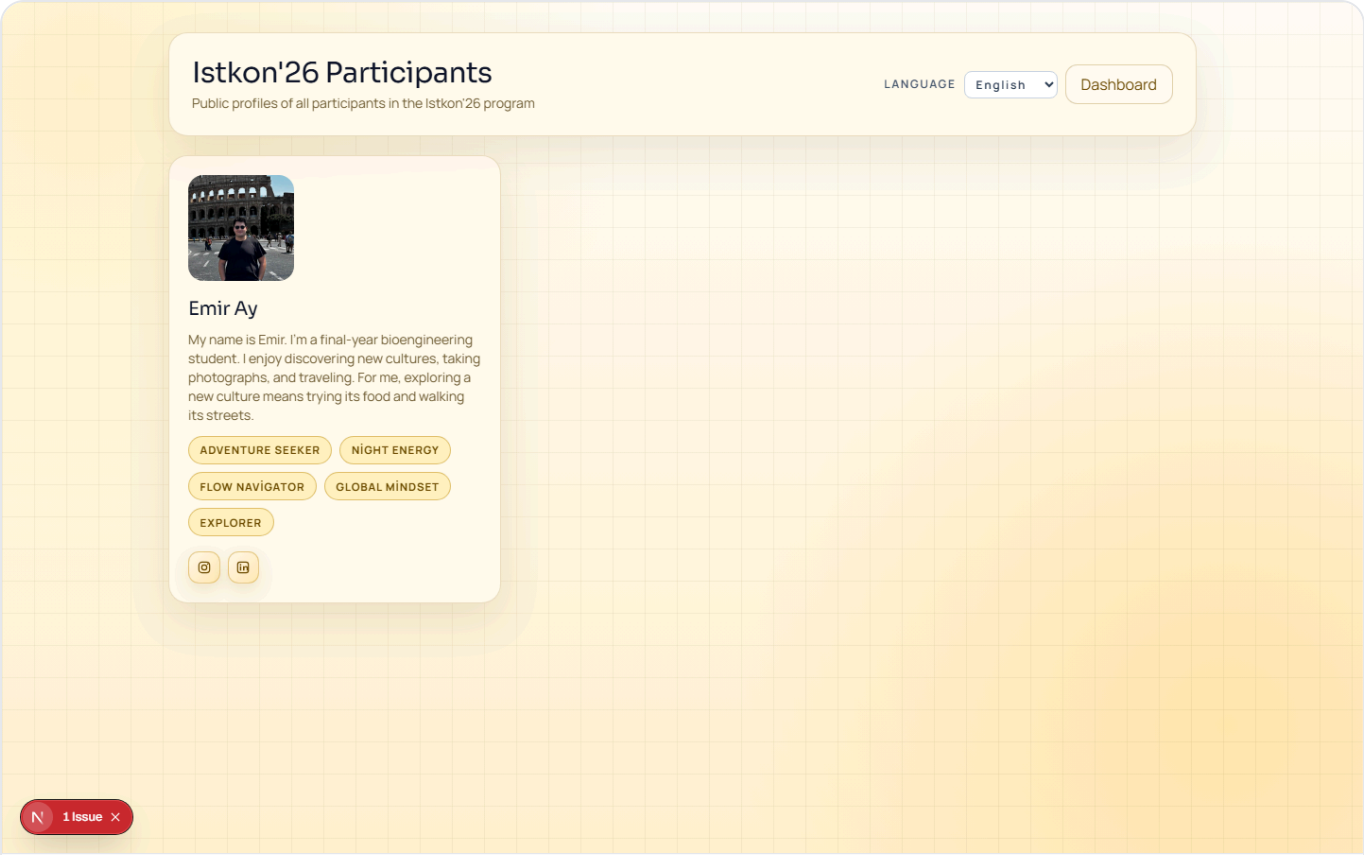
3. Dashboard Workflow



Dashboard with public profile editor, private answers entry point, and buddy overview.



Private answers modal used for Likert and forced-choice matching inputs.



Participants directory highlighting public bios, tags, and social links.

4. How To Use BuddyMatcher

1. Open BuddyMatcher and choose English from the language selector.
2. Register a participant account or log in with existing credentials.
3. Complete public profile details, including profile photo and short bio.
4. Open the private answers modal and complete all survey questions.
5. Save profile and answers to update compatibility calculations.
6. Check the dashboard to view your assigned buddy when matching is available.
7. Browse the participants directory to explore other public profiles.
8. Admins can import CSV data and trigger matching from the admin panel.

Operational Tip

For best matching quality, encourage participants to write clear bios in English and answer private questions consistently.

5. Matching Algorithm (Detailed)

BuddyMatcher uses a two-step approach: first it calculates pairwise compatibility scores between Turkish and German participants, then it computes a global optimal assignment using the Hungarian algorithm.

5.1 Pair constraints

- Only TR-DE pairs are allowed.
- TR and DE participant counts must be equal for one-to-one matching.
- If constraints are not met, matching is blocked.

5.2 Compatibility score per pair (0-100)

Weighted survey dimensions are normalized by absolute difference and contribute up to 76 points.

Dimension	Weight
Social	18
Openness	16
Flexibility	14
Structure	10
Party style	14
Travel style	12
Communication	16

Additional score components:

- Openness bridge score: up to +8 points.
- Night rhythm similarity: up to +4 points.
- Travel-after-program alignment: +4 if aligned, +0 otherwise.
- Forced-choice similarity: up to +8 points (fallback 4.4 for legacy profiles).

Forced-choice internal weighting:

Forced-choice block	Weight inside forced-choice similarity
Planning style	28%
Buddy priority	28%
Ideal activity	26%
Time style	18%

```
finalScore = clamp(0, 100,
  coreWeightedScore(0..76)
  + opennessBridge(0..8)
  + rhythmSimilarity(0..4)
  + travelAlignment(0 or 4)
  + forcedChoiceComponent(0..8)
)
```

5.3 Global assignment step (Hungarian)

- Build a cost matrix with cost = 100 - compatibilityScore.
- Run Hungarian optimization to minimize total cost across all pairs.
- This produces a globally optimal matching, not a greedy local choice.
- For each final pair, BuddyMatcher stores score and an explanation string.

6. Questions Used for Matching

Likert items use a 1-5 scale: 1 = Strongly disagree, 5 = Strongly agree.

1. Social Energy & Interaction

1. I easily start conversations with new people.
2. I enjoy being in large social groups.
3. I feel energized after social interactions.
4. I prefer spending time alone rather than with a group.
5. I actively participate in group activities.
6. I like meeting new people from different backgrounds.
7. I feel comfortable being the center of attention.
8. I usually wait for others to approach me first.

2. Openness & Cultural Curiosity

1. I am excited to experience new cultures.
2. I enjoy trying unfamiliar foods.
3. I like learning different perspectives.
4. I feel uncomfortable in unfamiliar environments.
5. I adapt quickly to new situations.
6. I enjoy stepping outside my comfort zone.
7. I am curious about how people live in other countries.

3. Planning & Lifestyle

1. I prefer having a clear plan for the day.
2. I get stressed when plans change suddenly.
3. I like organizing activities in advance.
4. I enjoy spontaneous decisions.
5. Being on time is very important to me.
6. I prefer structured schedules over flexible ones.
7. I can easily adjust to unexpected changes.

4. Social Activities & Party Style

1. I enjoy nightlife and parties.
2. I like high-energy social environments.
3. I prefer calm and quiet activities.
4. I enjoy dancing or active social events.
5. I feel comfortable in loud environments.
6. I prefer meaningful conversations over parties.

5. Daily Rhythm

1. I am more productive in the morning.
2. I enjoy staying up late.
3. I prefer starting the day early.
4. I feel active during late hours.

6. Travel & Exploration

1. I prefer fast-paced travel schedules.
2. I enjoy exploring multiple places in one day.
3. I prefer relaxed travel with fewer activities.

7. Communication & Comfort

1. I feel confident communicating in English.
2. I enjoy deep conversations with new people.
3. I can maintain conversations easily.
4. I prefer listening rather than speaking.

Travel preference (yes/no)

- I want to travel after the program. (Yes / No)

8A. Which describes you better?

- I plan ahead but can be flexible
- I am spontaneous but can plan when needed
- I prefer clear structure and detailed plans
- I go with the flow and adapt on the move

8B. Which matters more in a buddy?

- Fun and social
- Calm and reliable
- Curious and open to deep conversations
- Action-oriented and motivated

8C. Ideal activity choice

- Party / Social
- Cultural / Museum
- Mixed
- Outdoor / Nature

- Food / Coffee / City walk

8D. Preferred time style

- Early sleeper / early riser
- Late sleeper / night active
- Balanced day-night rhythm
- Late morning starter